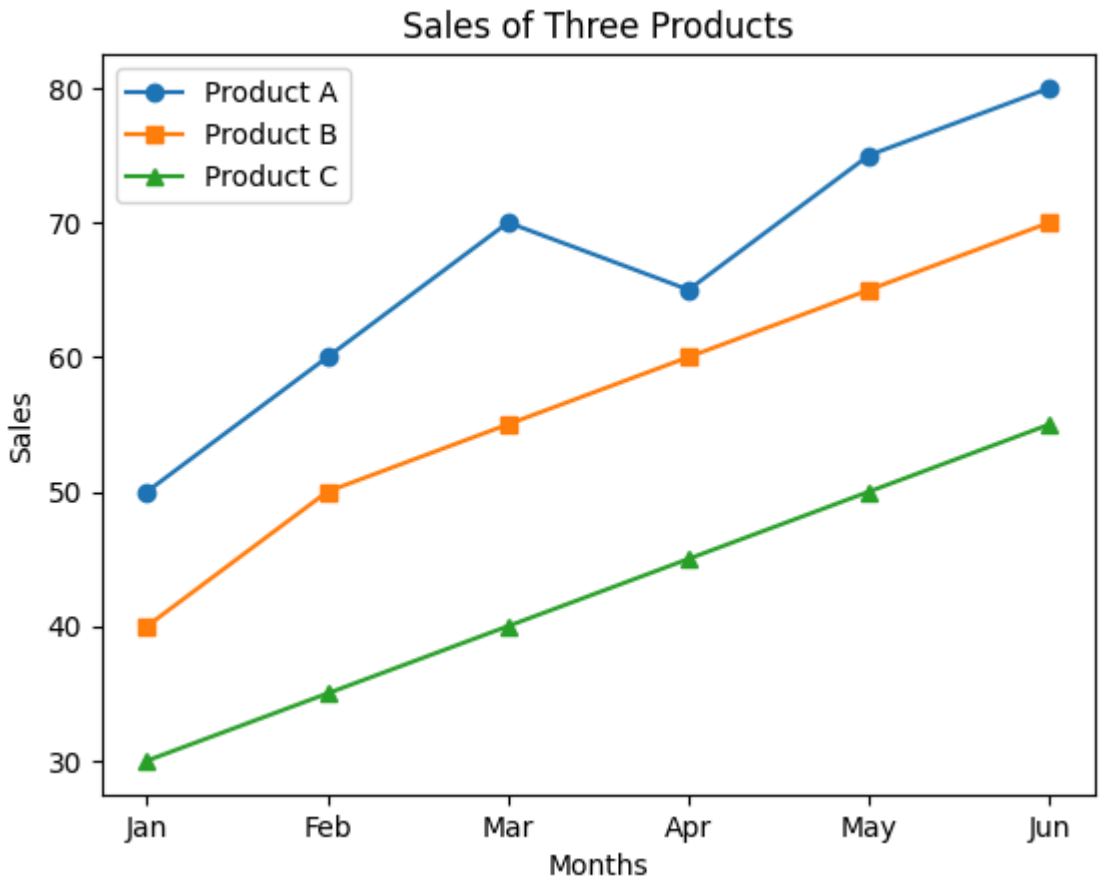
In [1]:
```
# Import matplotlib library
import matplotlib.pyplot as plt

# Months of the year
months = ["Jan","Feb","Mar","Apr","May","Jun","Jul","Aug","Sep","Oct","Nov","Dec

# Sales data for each month
sales = [120,140,130,150,170,160,180,200,190,210,220,230]

# Create line chart
plt.plot(months, sales, color="blue", marker="o")

# Add title and labels
plt.title("Monthly Sales of Company")
plt.xlabel("Months")
plt.ylabel("Sales")

# Display the chart
plt.show()
```

In [3]:
```
# Import matplotlib
import matplotlib.pyplot as plt

# Months
months = ["Jan","Feb","Mar","Apr","May","Jun"]

# Sales data of products
product_A = [50,60,70,65,75,80]
product_B = [40,50,55,60,65,70]
product_C = [30,35,40,45,50,55]
```

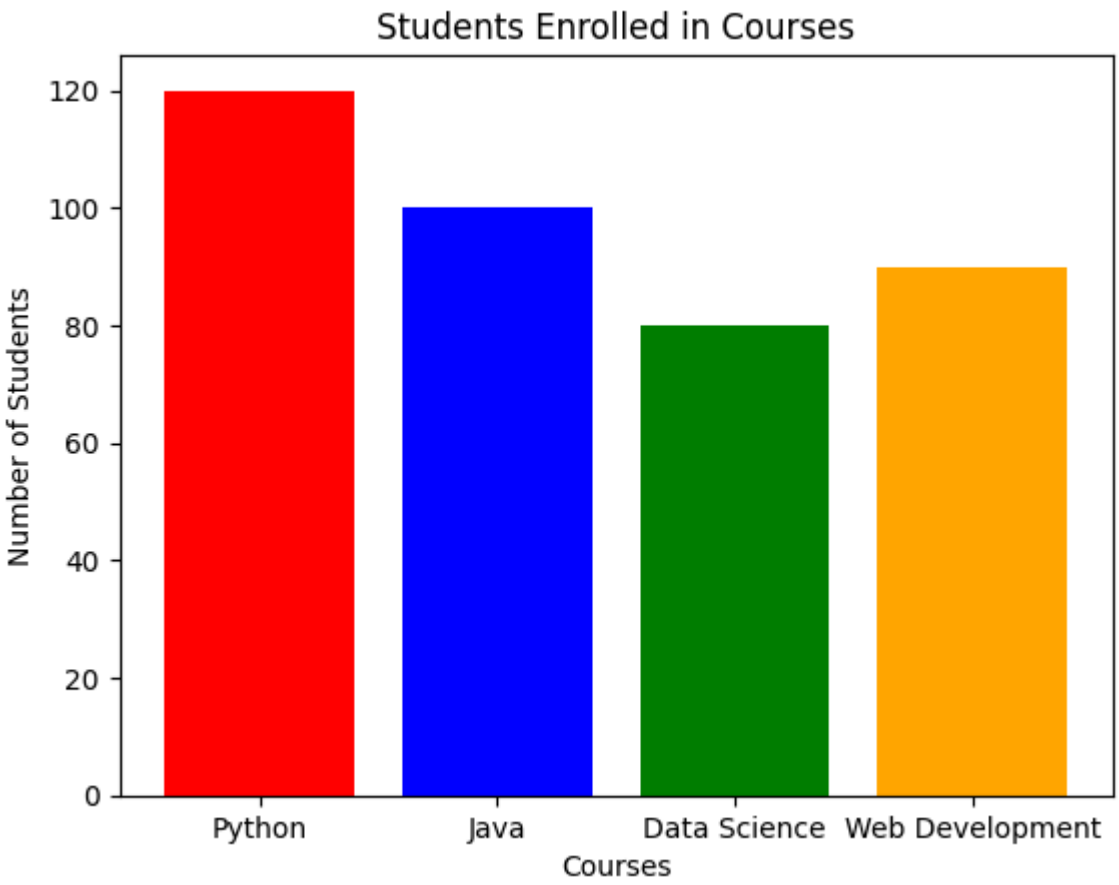
```
# Plot multiple lines
plt.plot(months, product_A, label="Product A", marker="o")
plt.plot(months, product_B, label="Product B", marker="s")
plt.plot(months, product_C, label="Product C", marker="^")

# Labels and title
plt.title("Sales of Three Products")
plt.xlabel("Months")
plt.ylabel("Sales")

# Show legend
plt.legend()

plt.show()
```



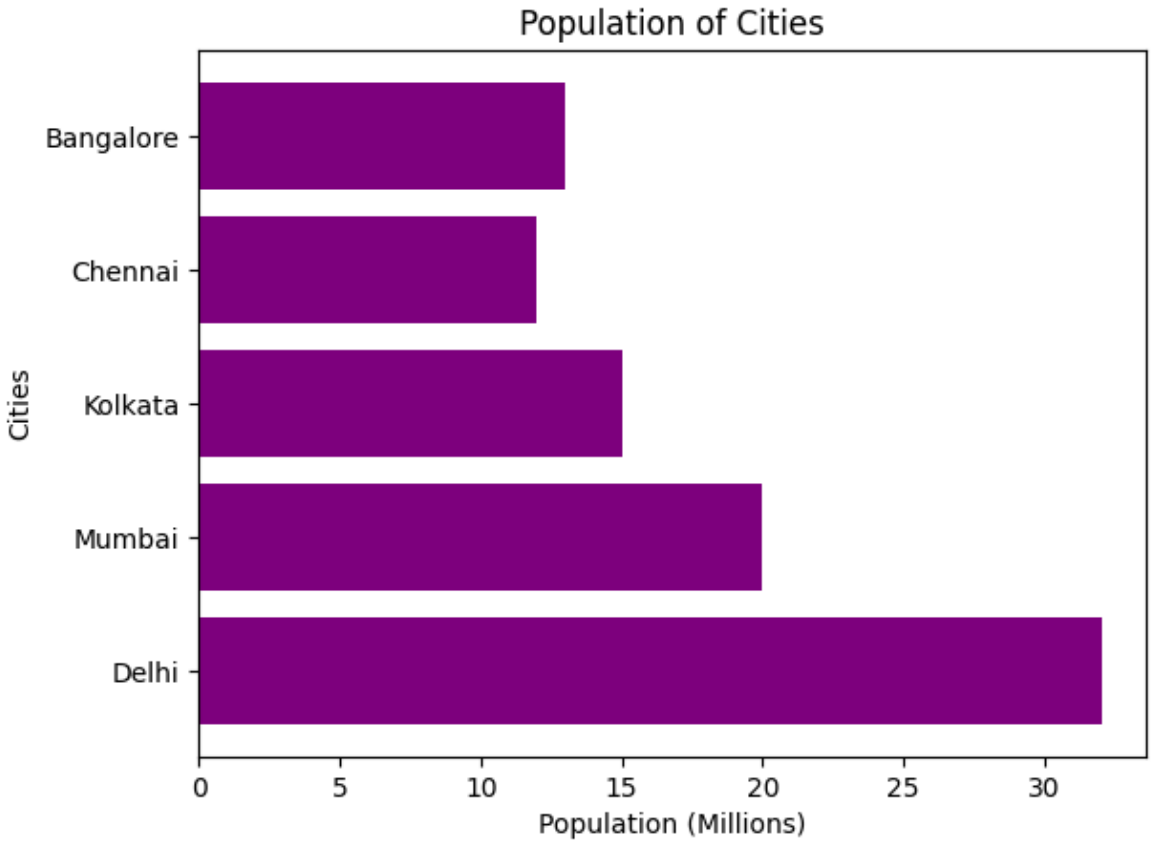
In [4]:

```
# Import matplotlib
import matplotlib.pyplot as plt

# Course names
courses = ["Python","Java","Data Science","Web Development"]

# Number of students
students = [120,100,80,90]

# Create bar chart
plt.bar(courses, students, color=["red","blue","green","orange"])

# Labels
plt.title("Students Enrolled in Courses")
plt.xlabel("Courses")
plt.ylabel("Number of Students")
```

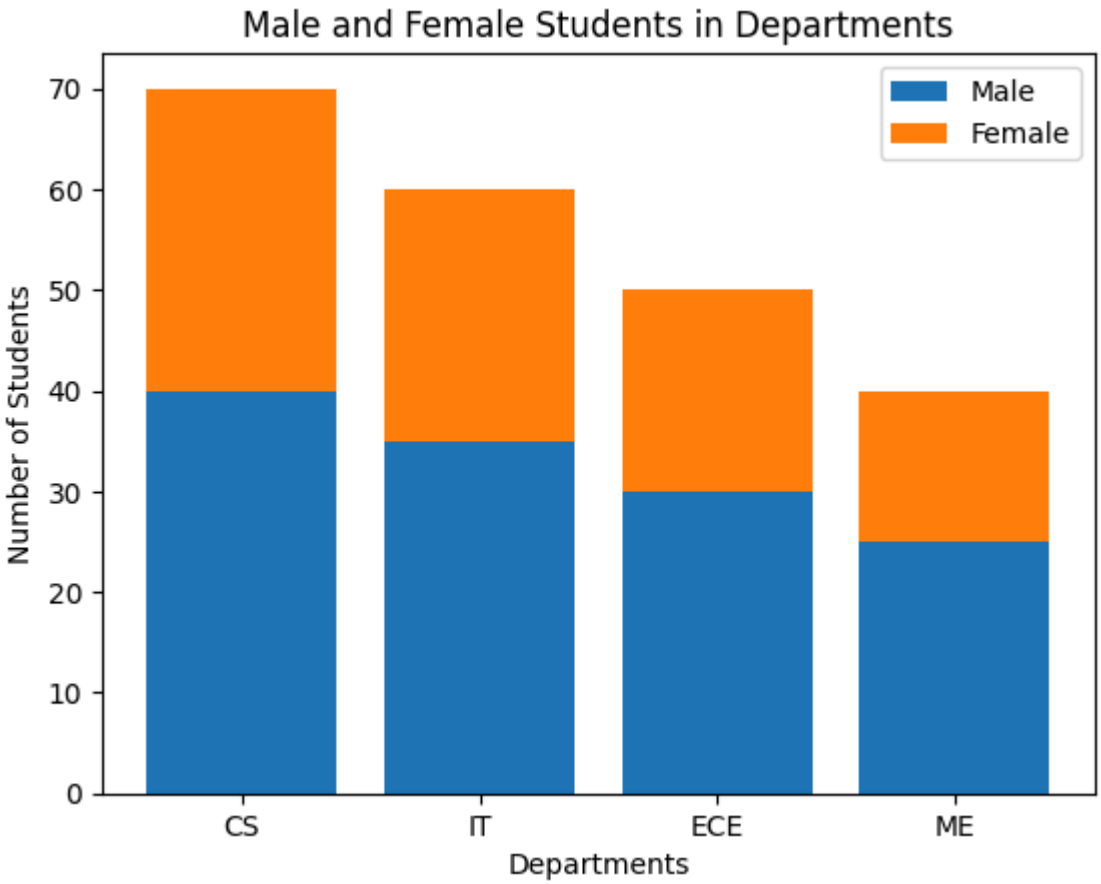
```
plt.show()
```



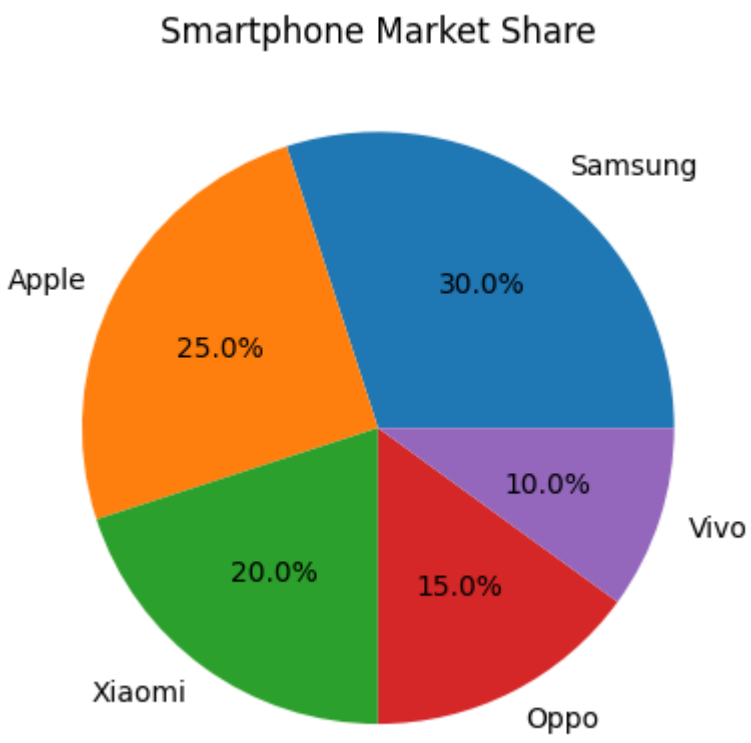
In [5]:
```
# Import matplotlib
import matplotlib.pyplot as plt

# Cities
cities = ["Delhi","Mumbai","Kolkata","Chennai","Bangalore"]

# Population data
population = [32,20,15,12,13]

# Create horizontal bar chart
plt.barh(cities, population, color="purple")

# Labels
plt.title("Population of Cities")
plt.xlabel("Population (Millions)")
plt.ylabel("Cities")

plt.show()
```



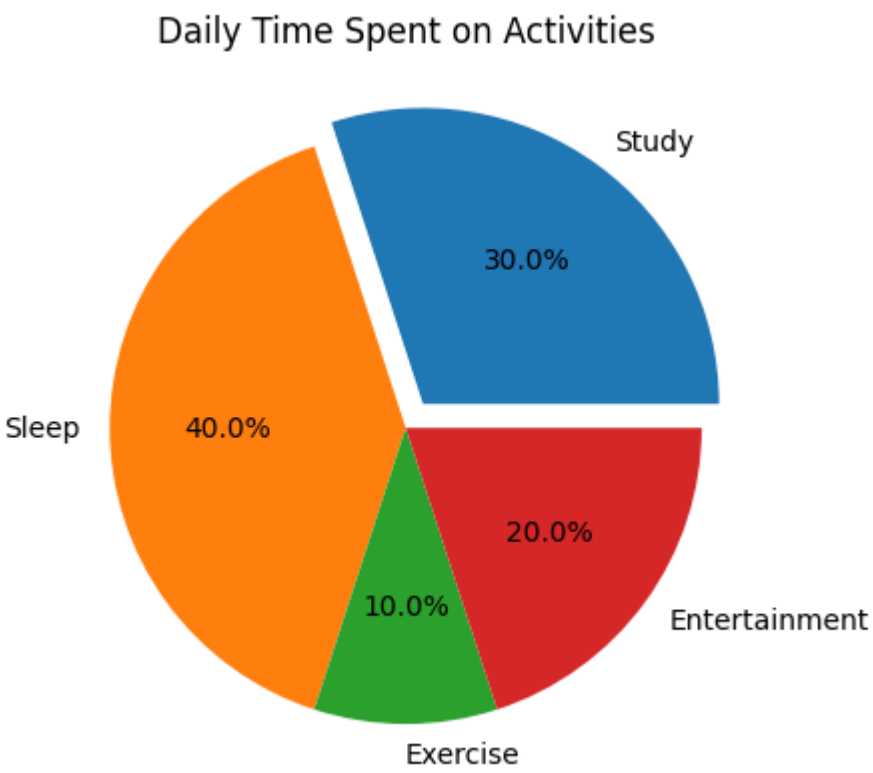
In [6]:
```
# Import matplotlib
import matplotlib.pyplot as plt

# Departments
departments = ["CS","IT","ECE","ME"]

# Male and Female students
male = [40,35,30,25]
female = [30,25,20,15]

# Plot stacked bars
plt.bar(departments, male, label="Male")
plt.bar(departments, female, bottom=male, label="Female")

# Labels
plt.title("Male and Female Students in Departments")
plt.xlabel("Departments")
plt.ylabel("Number of Students")

plt.legend()

plt.show()
```



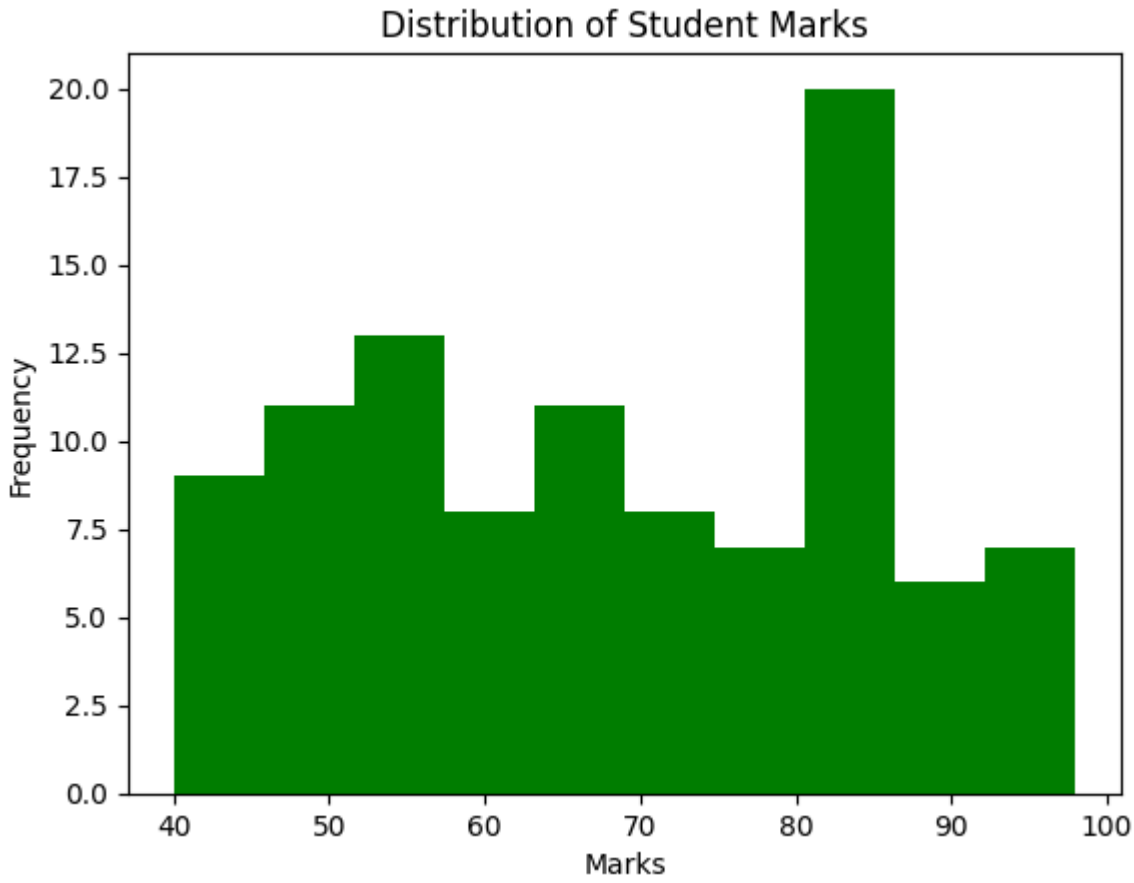
In [7]:
```
# Import matplotlib
import matplotlib.pyplot as plt

# Smartphone brands
brands = ["Samsung","Apple","Xiaomi","Oppo","Vivo"]

# Market share percentage
share = [30,25,20,15,10]

# Create pie chart
plt.pie(share, labels=brands, autopct="%1.1f%%")

# Title
plt.title("Smartphone Market Share")

plt.show()
```



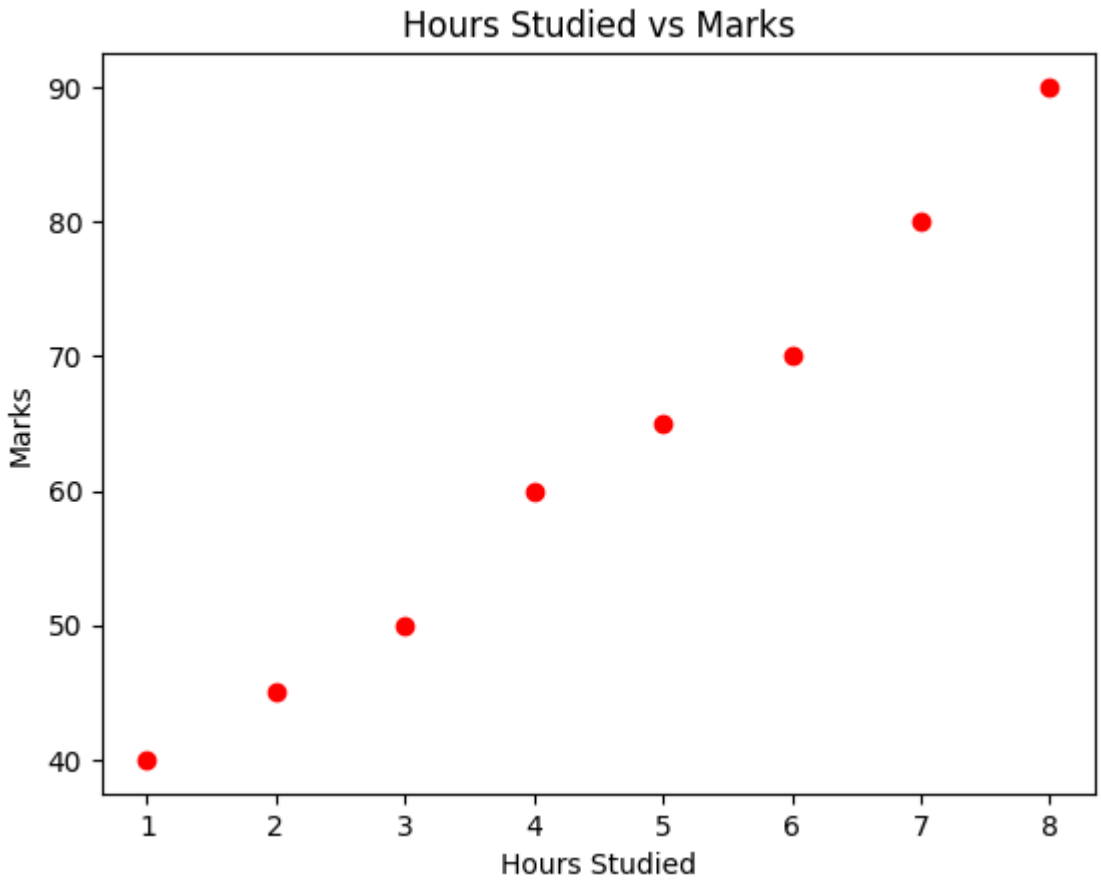
In [8]:
```
# Import matplotlib
import matplotlib.pyplot as plt

# Activities
activities = ["Study","Sleep","Exercise","Entertainment"]

# Time spent
time = [6,8,2,4]

# Explode first slice
explode = [0.1,0,0,0]

# Plot exploded pie chart
plt.pie(time, labels=activities, explode=explode, autopct="%1.1f%%")

plt.title("Daily Time Spent on Activities")

plt.show()
```



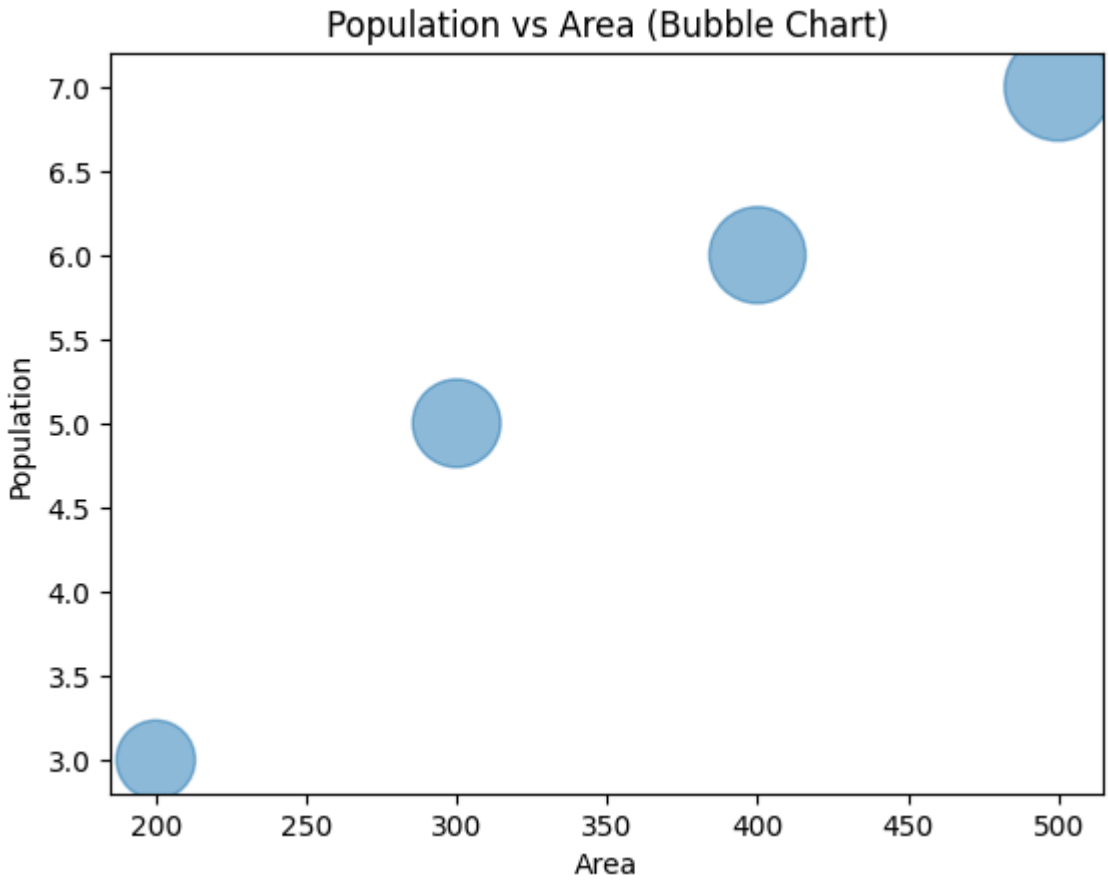
In [9]:
```
# Import libraries
import matplotlib.pyplot as plt
import numpy as np

# Generate random marks of 100 students
marks = np.random.randint(40,100,100)

# Create histogram
plt.hist(marks, bins=10, color="green")

# Labels
plt.title("Distribution of Student Marks")
plt.xlabel("Marks")
plt.ylabel("Frequency")

plt.show()
```



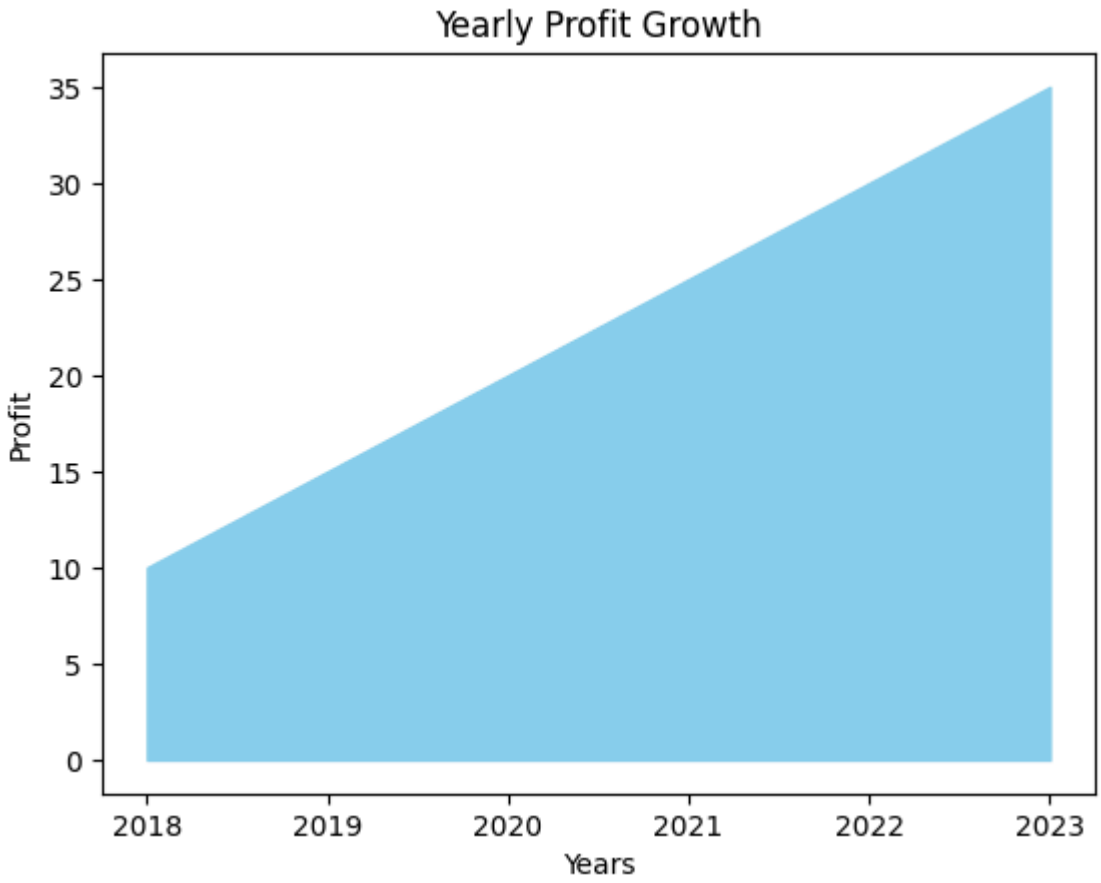
In [10]:
```
# Import matplotlib
import matplotlib.pyplot as plt

# Study hours
hours = [1,2,3,4,5,6,7,8]

# Marks obtained
marks = [40,45,50,60,65,70,80,90]

# Create scatter plot
plt.scatter(hours, marks, color="red")

# Labels
plt.title("Hours Studied vs Marks")
plt.xlabel("Hours Studied")
plt.ylabel("Marks")

plt.show()
```



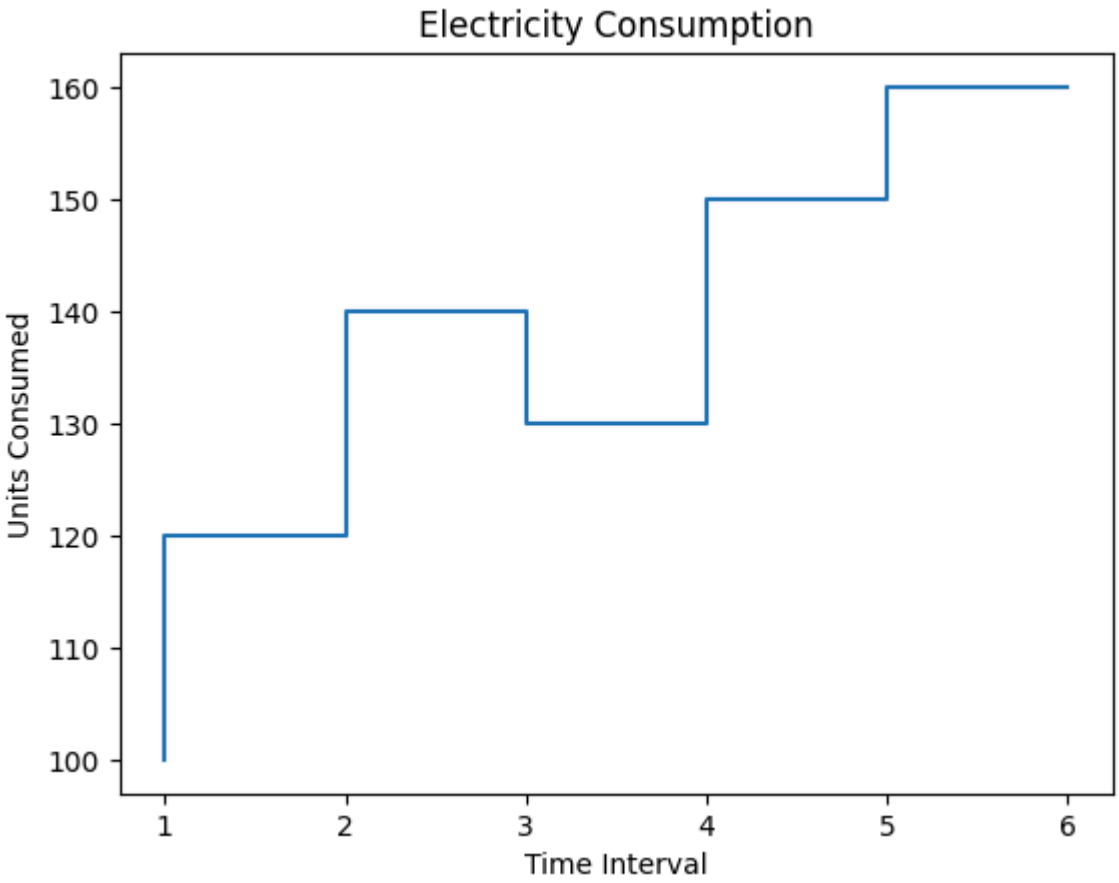
In [11]:
```
# Import matplotlib
import matplotlib.pyplot as plt

# City data
population = [5,7,3,6]
area = [300,500,200,400]
gdp = [1000,1500,800,1200]

# Bubble chart
plt.scatter(area, population, s=gdp, alpha=0.5)

# Labels
plt.title("Population vs Area (Bubble Chart)")
plt.xlabel("Area")
plt.ylabel("Population")

plt.show()
```



In [12]:
```
# Import matplotlib
import matplotlib.pyplot as plt

# Years
years = [2018,2019,2020,2021,2022,2023]

# Profit data
profit = [10,15,20,25,30,35]

# Area chart
plt.fill_between(years, profit, color="skyblue")

# Labels
plt.title("Yearly Profit Growth")
plt.xlabel("Years")
plt.ylabel("Profit")

plt.show()
```

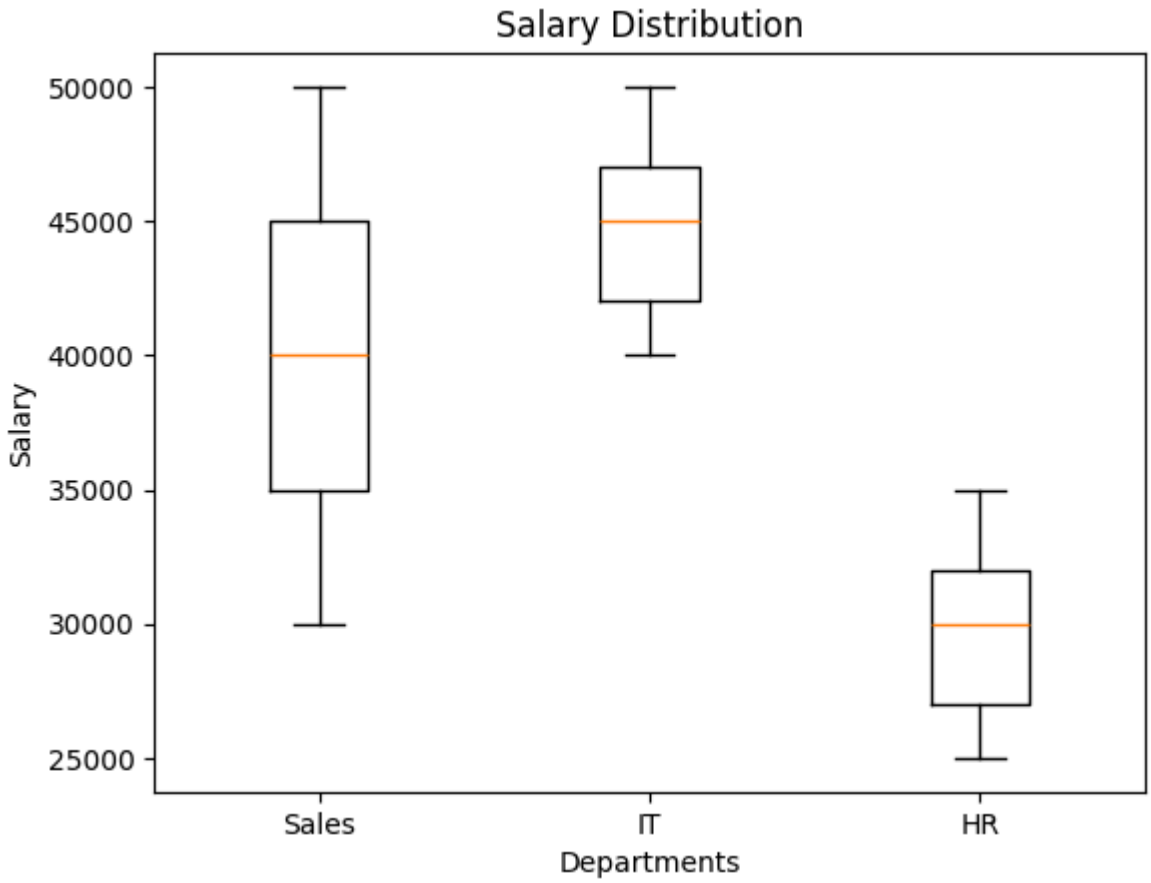


In [13]:
```
# Import matplotlib
import matplotlib.pyplot as plt

# Time intervals
time = [1,2,3,4,5,6]

# Electricity consumption
consumption = [100,120,140,130,150,160]

# Step chart
plt.step(time, consumption)

# Labels
plt.title("Electricity Consumption")
plt.xlabel("Time Interval")
plt.ylabel("Units Consumed")

plt.show()
```

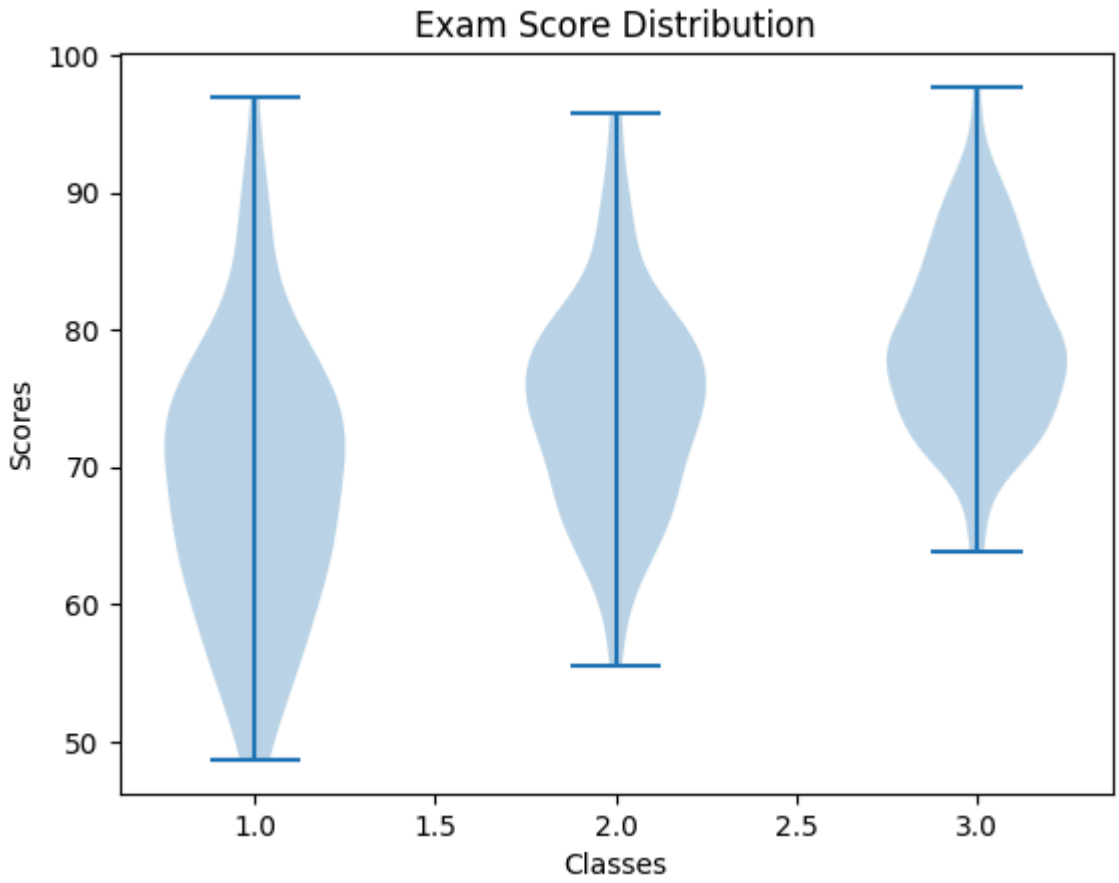


In [14]:
```
# Import matplotlib
import matplotlib.pyplot as plt

# Salary data
sales = [30000,35000,40000,45000,50000]
it = [40000,42000,45000,47000,50000]
hr = [25000,27000,30000,32000,35000]

# Create box plot
plt.boxplot([sales,it,hr], labels=["Sales","IT","HR"])

# Labels
plt.title("Salary Distribution")
plt.xlabel("Departments")
plt.ylabel("Salary")

plt.show()
```

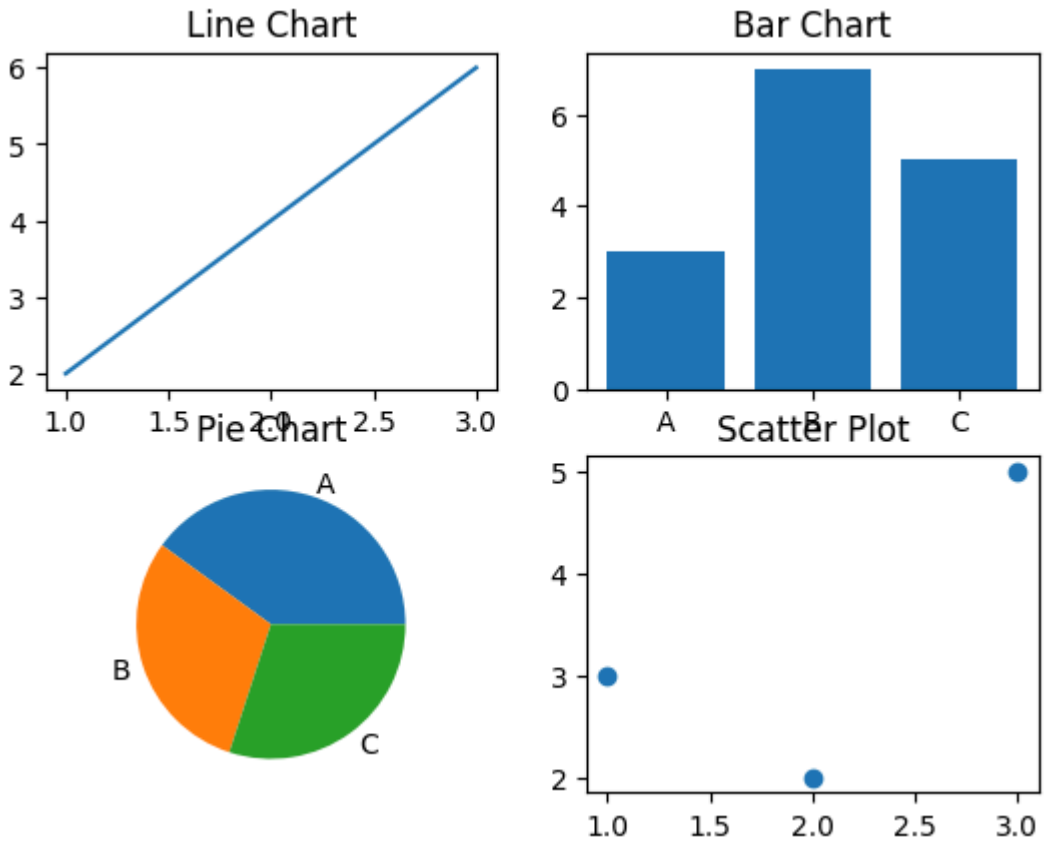
```
C:\Users\rathi\AppData\Local\Temp\ipykernel_5260\1253288111.py:10: MatplotlibDepr
ecationWarning: The 'labels' parameter of boxplot() has been renamed 'tick_label
s' since Matplotlib 3.9; support for the old name will be dropped in 3.11.
  plt.boxplot([sales,it,hr], labels=["Sales","IT","HR"])
```



In [15]:
```
# Import libraries
import matplotlib.pyplot as plt
import numpy as np

# Generate exam scores
classA = np.random.normal(70,10,100)
classB = np.random.normal(75,8,100)
classC = np.random.normal(80,7,100)

# Create violin plot
plt.violinplot([classA,classB,classC])

# Labels
plt.title("Exam Score Distribution")
plt.xlabel("Classes")
plt.ylabel("Scores")

plt.show()
```



In [16]:
```
# Import matplotlib
import matplotlib.pyplot as plt

# Create subplots
fig, ax = plt.subplots(2,2)

# Line chart
ax[0,0].plot([1,2,3],[2,4,6])
ax[0,0].set_title("Line Chart")

# Bar chart
ax[0,1].bar(["A","B","C"],[3,7,5])
ax[0,1].set_title("Bar Chart")

# Pie chart
ax[1,0].pie([40,30,30], labels=["A","B","C"])
ax[1,0].set_title("Pie Chart")

# Scatter plot
ax[1,1].scatter([1,2,3],[3,2,5])
ax[1,1].set_title("Scatter Plot")

plt.show()
```


Line Chart
6
5
4
3
2
1.0
1.5
2.0
2.5
3.0
Bar Chart
6
4
2
0
A
B
C
Pie Chart
A
B
C
Scatter Plot
5
4
3
2
1.0
1.5
2.0
2.5
3.0

In [ ]: